



**Politecnico
di Torino**

Department
of Electronics and
Telecommunications



**Politecnico
di Torino**

Department of Environment,
Land and Infrastructure
Engineering

ICT FOR GEOMATICS: NAVIGATION AND MAPS

01QWUBH

Lab Report

Authors:

Lorenzo Cavallaro - s346742

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1 Lab 1: Raw GNSS Measurements in Android Smartphones

1.1 Introduction

This lab session focused on the analysis of raw GNSS measurements collected by using the GNSSLogger app running on an Android mobile device. First we used demo datasets to then compare the results with data collected directly from our own Android devices.

1.2 Data Processing with MATLAB

The data were processed by using a given open source MATLAB script that implements the following functionalities:

- Parsing and filtering raw GNSS measurements retrieved from the log file.
- Downloading satellite ephemeris data from NASA's CDDIS service to support satellite positioning.
- Applying corrections to the reception times to account for clock bias and system offsets.
- Computing key data as pseudoranges, pseudorange rates, and carrier-to-noise ratio (C/N_0).
- Plotting the evaluated results to understand the behavior of the geopositioning system over time.

1.3 Raw Data Parsing and Filtering

The first step of the analysis was to filter the raw GNSS data contained in the log file, saved in .txt format. This file includes information about the signals received by the GNSS satellites, such as transmission and reception time, satellite identifiers, and signal quality indicators. The filter focused on excluding incomplete, inconsistent, or irrelevant data, by applying the following conditions:

- Discard the measurements where the receiver clock bias (FullBiasNanos) is equal to 0, because this signifies an invalid or uninitialized measurement.
- Consider only signals coming from GPS satellites.
- Keep only the measurements in which the code lock was achieved and the Time of Week (TOW) was successfully decoded.

1.4 Time Corrections

Raw GNSS data collected by mobile devices contains various biases and offsets that must be corrected in order to perform a meaningful PVT solution. Therefore, before computing pseudoranges, a time correction was applied by removing from the reception time, measured in nanoseconds, the full bias with the satellite's clock:

$$t_{\text{Rx, nanos}} = \text{TimeNanos} - \text{FullBiasNanos} \quad (1)$$

To convert the reception time in GPS time, it is necessary to extract the GPS week and compute the time of week (TOW):

$$t_{\text{Rx, nanos}} = t_{\text{Rx, nanos}} - (\text{Week Number} \times 604800 \times 10^9) \quad (2)$$

Lastly, smaller corrections are needed to account for receiver's clock biases caused by quartz clock inaccuracy, and offsets caused by relativistic effects:

$$t_{\text{Rx, sec}} = \frac{t_{\text{Rx, nanos}} - \text{TimeOffsetNanos} - \text{BiasNanos}}{10^9} \quad (3)$$

Once the reception time is fully corrected, it is used with the signal transmission time, contained in the raw data, to evaluate the time of travel.

$$\text{prSeconds} = t_{\text{Rx, sec}} - t_{\text{Tx, sec}} \quad (4)$$

In case of week rollover, the reception time falls in the next week respect to the transmission, so the evaluated travel time will have a bigger value than expected and must be corrected by removing a Δt .

Since we know the satellite's position, retrieved from the NASA Ephemeris data, and the transmission time and speed (signals travel at speed of light), it is possible to estimate how far the receiver is from the satellite. To solve the position of the receiver, the combination of at least 4 satellite's signals is required.

1.5 Pseudoranges

Pseudoranges measure how far the receiver is from the satellite that has sent the signal. To calculate this distance is necessary to multiply the travel time with the speed of light, and an estimation in meters is obtained as result.

Figure 1a contains 3 different plots related to pseudoranges computations:

- **Pseudoranges vs Time**, shows the trend of the pseudoranges evaluated for each satellite. The measures are taken in meters.
- **Pseudoranges change from initial value vs Time**, illustrates how the pseudorange values differ from their initial value. Those satellites which the

pseudorange increases are moving away from the receiver, on the contrary those which the pseudoranges are shrinking are moving towards.

- **Hardware Clock Discontinuity**, red points coincide to timestamp where the receiver's clock resets, or jumps and breaks the continuity of the time measures.

Figure 1b shows in meters per second how fast the pseudoranges are changing over time. This velocity can be interpreted as the speed of motion of the satellite orbiting around the earth, since in this scale the receiver's speed is negligible. To each satellite are associated two lines having the same path, but one is more noisy than the other. The most steady one represents the pseudorange rates estimated in the raw GNSS data, on the other hand the colored line is obtained by evaluating, for each point, the difference between the actual pseudorange and the previous.

Notice that both figures show the same pseudorange discontinuity points, meaning that at those timestamps no signal has been received from the satellite.

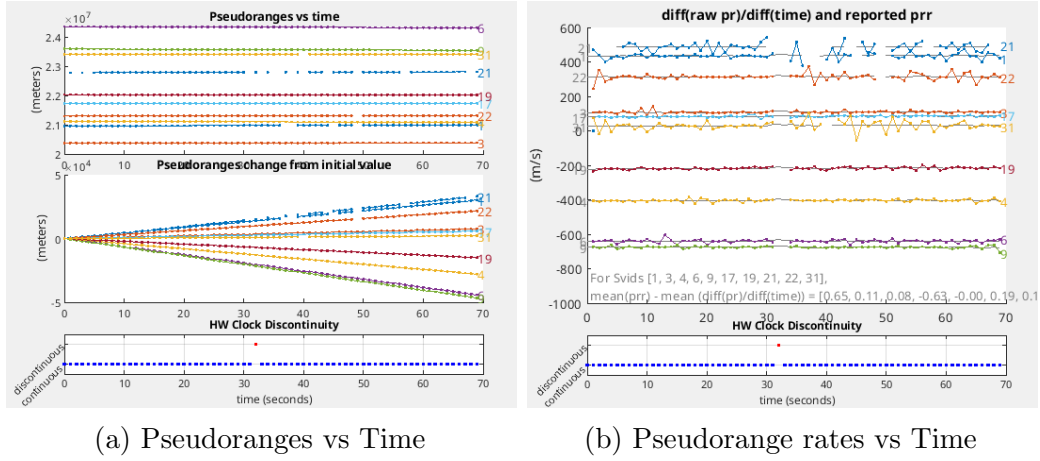
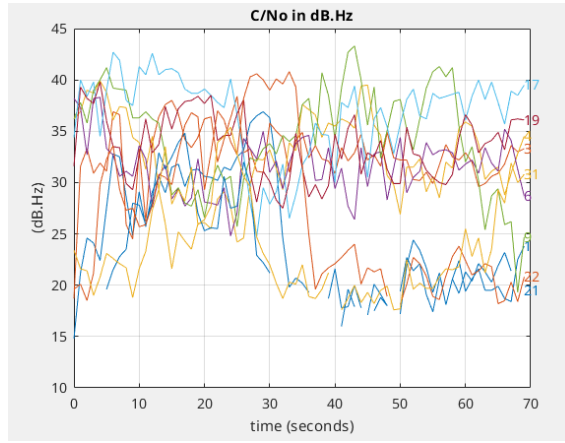


Figure 1: Demo dataset: dataset_a/gnss_log_2021_10_11_12_00_32.txt

1.6 Carrier-to-Noise Density Ratio

C/N_0 measures the signal strength and quality by evaluating the ratio between the carrier power (C) and the noise power density per Hz of bandwidth (N_0). Figure 2 illustrates how the signal quality changes over time for each satellite. Even if we would expect that the receiver gets stronger signals from nearer satellites, this behavior could be affected by possible disturbs during the signal transmission, as multipath or interference. Additionally, we can compare figure 2 with figure 1b to notice that weaker signals correspond to less accurate estimates, so to noisy lines for pseudorange rates. In the figure we can detect an overall reduction of signals quality, starting between the 30 and 40 epoch.



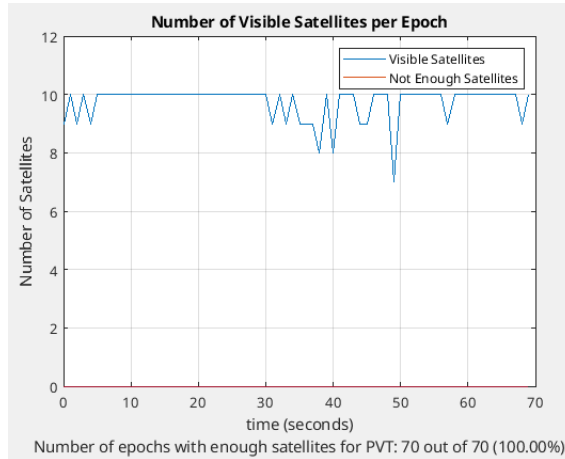
(a) C/N_0 vs Time

Figure 2: Demo dataset: dataset_a/gnss_log_2021_10_11_12_00_32.txt

1.7 Satellite Visibility and PVT Availability Analysis

Each satellite signal permits the computation of the pseudorange, so the distance between the satellite and the receiver. However, this computation is not enough to understand with accuracy the position of the user, since to solve PVT at a given timestamp we need to combine at least 4 pseudoranges, each evaluated from a different satellite.

To prepare the PVT computation, we have been asked to design a routine that counts the number of available satellites at each timestamp, and then estimate the percentage of time in which the number of visible satellites is sufficient. Figure 3) demonstrates that in the current demo dataset we have high satellites availability (100%), with a maximum value of 10 and a minimum of 7.



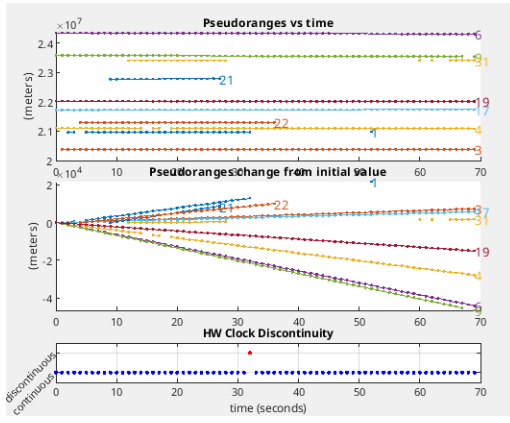
(a) Number of Visible Satellites vs Time

Figure 3: Demo dataset: dataset_a/gnss_log_2021_10_11_12_00_32.txt

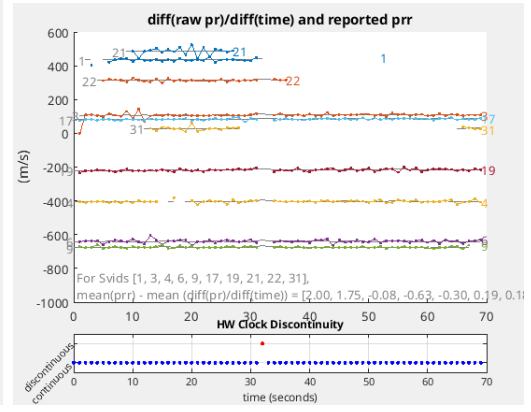
1.8 Impact of Signal Quality and Constellation Filters on PVT Solutions

We tested the impact of setting a signal quality filter that consider only those signals having C/N_0 grater than 24 db-Hz. The obtained plots show a drop in the satellites availability (see figure 4d), and as a consequence a reduction on the pseudorange computations (see figures 4a, 4b). In addition, since we are removing all the signals having low power and quality, figure 4c illustrates higher minimum values of C/N_0 than the previous test.

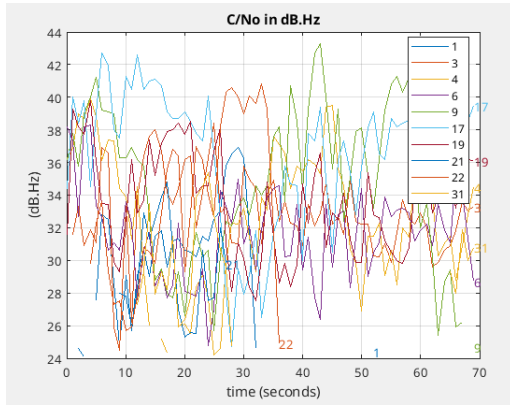
A second test was conducted to analyze the GNSS signals coming only from GALILEO satellites. Figure 4h demonstrates fewer visible satellites respect to the GPS (without filtering C/N_0). However, the number of available satellites never goes under 4, so PVT can be computed in 100% of the epochs. Moreover, we know that GALILEO satellites have an orbit altitude approximately of 23,222 *km*, slightly higher than GPS orbit of 20,200 *km*. This can be seen in figure 4e, where the pseudoranges have higher values than the previous test with GPS.



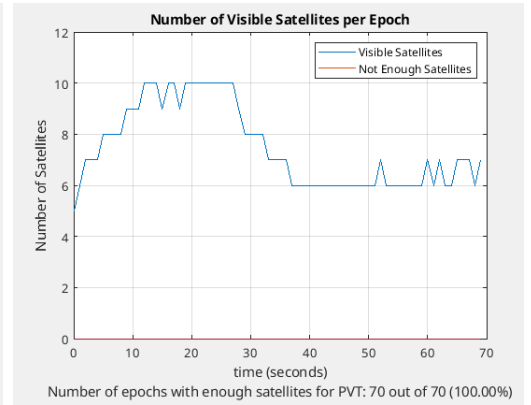
(a) Pseudoranges vs Time



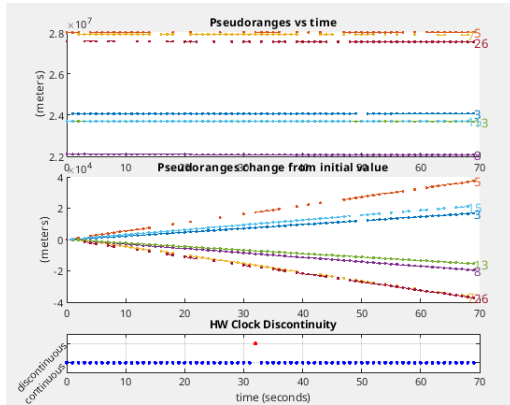
(b) Pseudorange rates vs Time



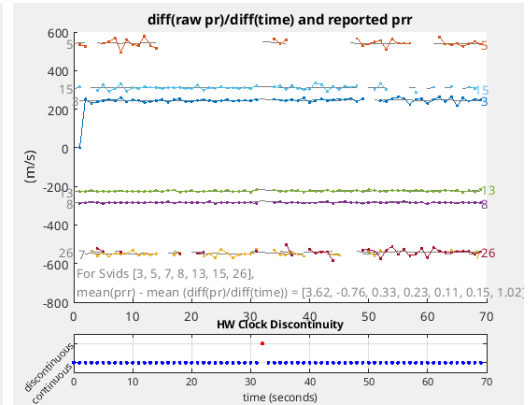
(c) C/N_0 vs Time



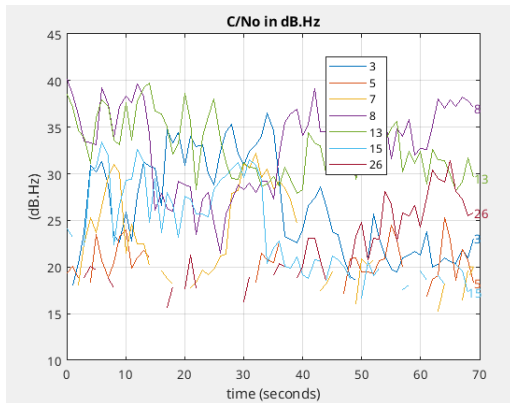
(d) Number of Visible Satellites vs Time



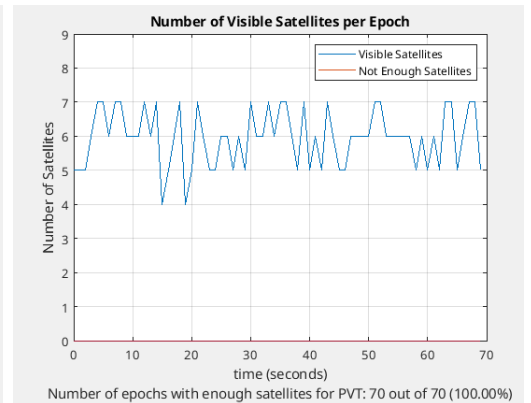
(e) Pseudoranges vs Time



(f) Pseudorange rates vs Time



(g) C/N_0 vs Time



(h) Number of Visible Satellites vs Time

Figure 4: Demo dataset: dataset_a/gnss_log_2021_10_11_12_00_32.txt

1.9 Analysis with GNSSLogger App Dataset

After having completed the previous analysis, we have been asked to perform the same investigations by using raw data collected through the GNSSLogger app on our personal Android smartphones.

The data used for this task were collected during my walk home, started from the Politecnico and concluded inside my bedroom. Compared to the demo dataset, this test shows better information about how daily geopositioning systems works while we are walking in the crowded streets of a city, and when we enter inside an indoor space. Data were collected over a period of approximately 15 minutes.

Figure 5 contains the resulting plots obtained by using the personal GNSS data, filtered only for the GPS constellation. During the path we see an high availability of satellites, demonstrating that in outdoor spaces the visibility is optimal. However, due to environmental factor, as the presence of high buildings and vegetation along the street, the overall signal power and strength experience an elevated variance. In addition, is interesting to notice the behavior of the geopositioning system in the last part of the test. In figure 5d, we see a critic drop in the amount of visible satellites, given that I entered inside my home. This drop is highlighted with the red points, in which each represents a timestamp where there are not enough visible satellites to solve PVT. As a consequence, we see from figure 5c that also C/N_0 plunges around the epoch 800, in coincidence with figure 5d.

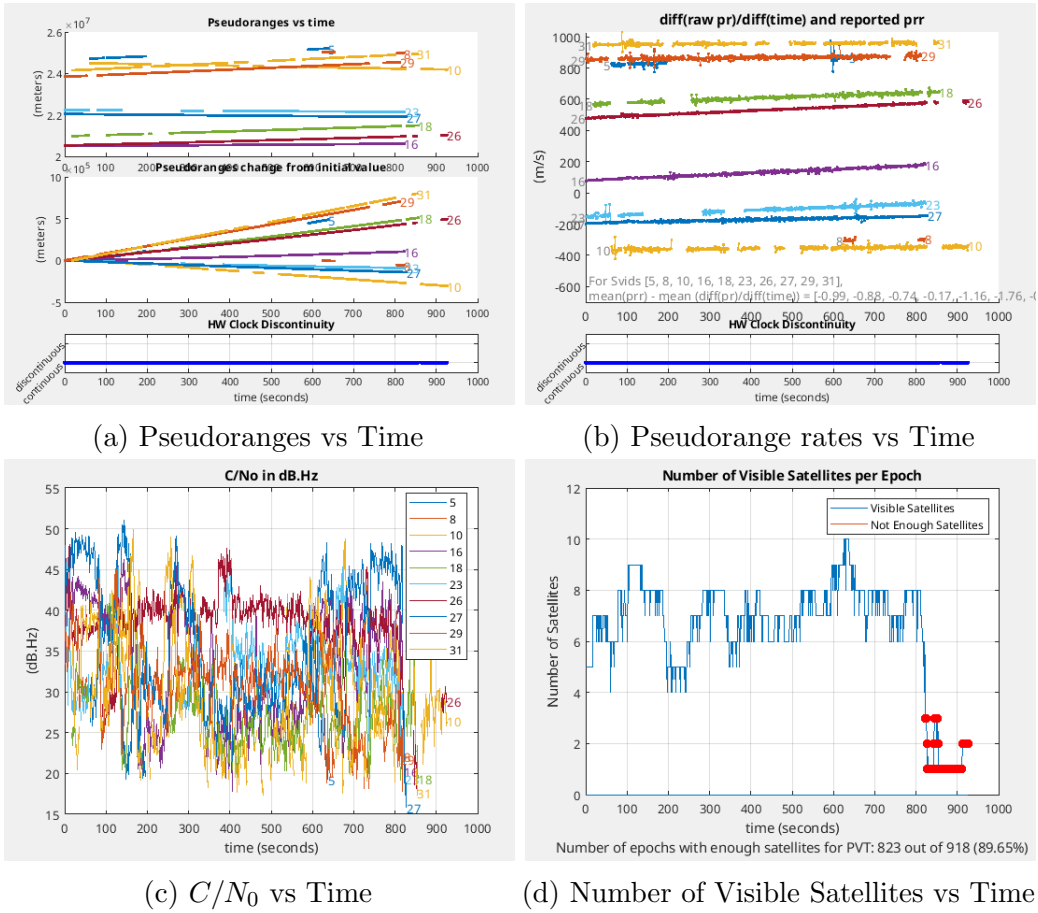


Figure 5: GNSSLogger app dataset: myLogs/gnss_log_2025_03_29_13_28_44.txt

1.10 Analyze Satellites Relative Positions and C/N_0 Relationships

The data collection was conducted on March 29th at 1:30 PM and lasted approximately 15 minutes. By using the GNSS Planning framework offered by Trimble, we study the relationship between the signal strength and the skyplot showing the visible GPS satellites over Turin in the same period of time (see Figure 6). Since the comparison is done with the previous test that did not use any signal strength filter, the skyplot is displayed with a cutoff angle equals to 0.

Figure 6a displays 11 visible satellites, in which 2 are near the zenith (exactly above the head of the receiver), and 8 are above an elevation angle of 15° . Lastly, figure 6b shows how the position of the satellites changed after 10 minutes, illustrating a good availability of 10 satellites, with 9 of those inside a cutoff angle of 15° .

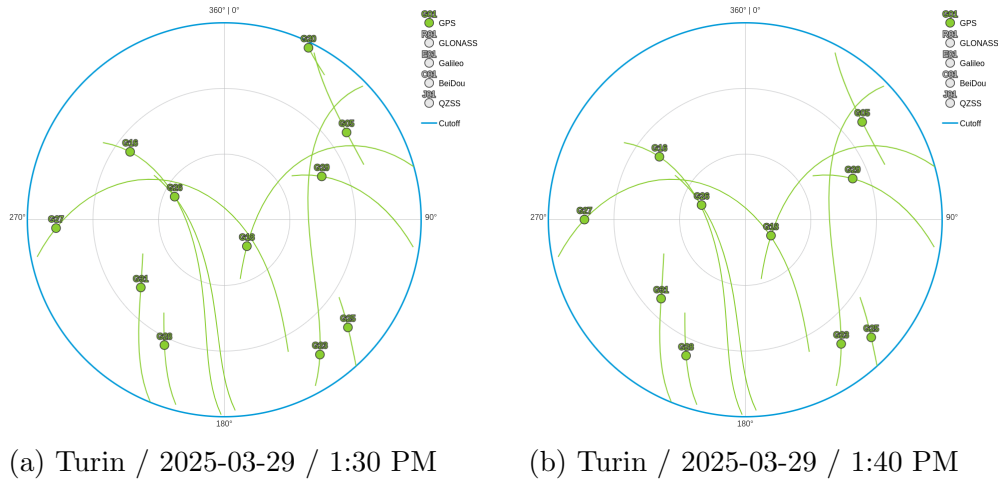


Figure 6: Skyplot of Visible Satellites without Cutoff Angle

In the skyplots the satellites with the Svids 5, 16, 18, 23, 26, 27, 29, 31 are in common with the C/N_0 plot. Figure 7 compares the signal strength of satellite 5 and satellite 26, to highlight how more consistent and strong is the signal coming from a satellite nearby the zenith.

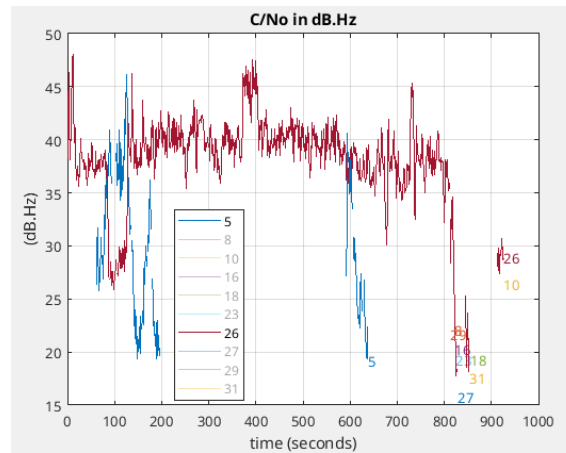


Figure 7: C/N_0 of Satellites 5 and 26

2 PVT Estimation

3 Other Labs